

The final exam of this course covers the following topics:

- Single discrete and continuous random variables including PMF, PDF, expected value, and variance of random variables. A summary of important random variables is already uploaded.
- Topics from the book "OpenIntro Statistics": 3.4.3 and 3.4.4
- Following topics from the text book:

8.3, 8.4 (before the Sampling Distribution of the Difference between Two Means), 8.5, and 8.6

9.1, 9.2, 9.3, 9.4, and 9.12

10.4, 10.5, 10.10 (excluding F-Test for Testing Variances in SAS), 10.11 (excluding the Geary's Test), and 10.12

11.1, 11.2, 11.3, and 11.5

12.1, 12.3, 12.5 (before the Inferences on Mean Response and Prediction), 12.6, 12.8, 12.9, and 12.12

13.1, 13.2, and 13.3

A formula sheet is uploaded that will be provided in the exam along with the required tables.

Students are supposed to know the definitions and formulas not mentioned in this list. Moreover, if any questions requires the use of a PMF/PDF, it will be provided in that question

Formula Sheet

$$s_p^2 = \frac{s_1^2(n_1 - 1) + s_2^2(n_2 - 1)}{n_1 + n_2 - 2}$$

$$v = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{(s_1^2/n_1)^2/(n_1 - 1) + (s_2^2/n_2)^2/(n_2 - 1)}$$

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2, \quad S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2, \quad S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

$$s^2 = \frac{SSE}{n-2} = \sum_{i=1}^n \frac{(y_i - \hat{y}_i)^2}{n-2} = \frac{S_{yy} - b_1 S_{xy}}{n-2}$$

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 :$$

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$$R^2_{\text{adj}} = 1 - \frac{SSE/(n-k-1)}{SST/(n-1)}$$

$$SSA = \sum_{j=1}^K N_j (\bar{Y}_{j.} - \bar{Y}_{..})^2$$

$$SSE = \sum_{j=1}^K \sum_{i=1}^{N_j} (y_{ij} - \bar{Y}_{j.})^2$$